RISK PORTFOLIO REPORTING

Automated risk portfolio reporting

Reporting Community

Reporting date 2026-09-30 | Peer group COMMUNITY

Synthetic demonstration report

This report combines a shared ingestion and data-quality platform with two analytical modules: Rating & Scoring and Loss Estimation. It is a runnable prototype built exclusively on fictional Sparkassen, portfolios and cashflows.

Institution scope	**Reporting Community**	Community view or single fictional institution.
Current EAD	**3,012,010,234 EUR**	Positive exposure at default in the reporting snapshot.
Exposure-weighted PD 12M	**1.78%**	Configured demonstration PD mapping, not a validated model.
Observed 12M default rate	**5.17%**	Backward-looking 12-month cohort.
Observed 12M recovery rate	**17.70%**	Realised recovery cashflows through the reporting date.
Observed 12M loss rate	**82.30%**	EAD at default minus realised recoveries.
Data quality score	**99.78%**	Valid records / received records across completed batches.
Data version	**RUN-20261111005418-7B456F2D**	Latest pipeline run identifier.

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

No real Sparkasse, customer, employee or portfolio is represented. Values are generated with a fixed seed and controlled data-quality defects for demonstration. The report is not regulatory reporting and must not be used for supervisory, accounting or customer decisions.

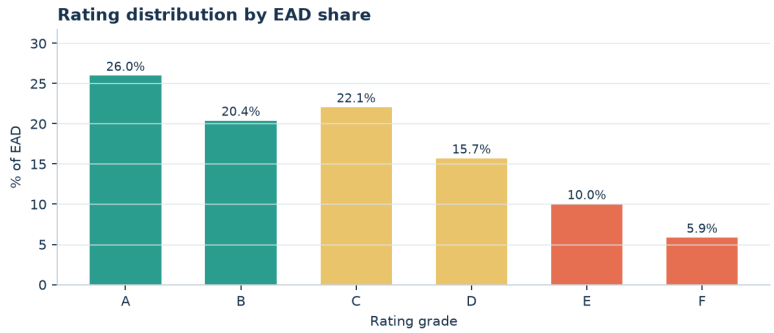
1. Executive overview

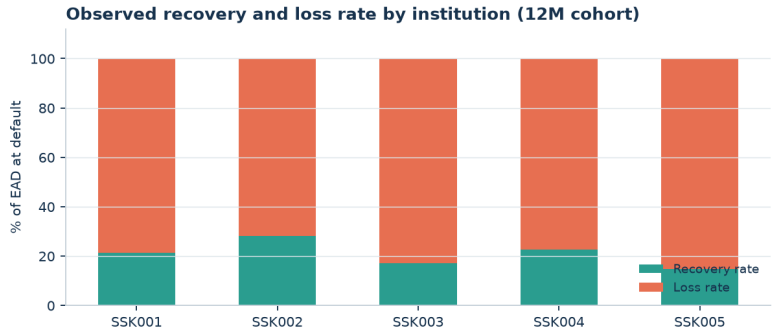
One reporting date, two analytical modules and a shared data foundation.

At 2026-09-30, the selected scope contains 2,095 exposures with total EAD of EUR 3,012,010,234. The exposure-weighted configured PD 12M is 1.78%. The observed 12-month default rate for the backward cohort is 5.17% (88 defaults on a base of 1,701 performing exposures). Observed recovery through the reporting date is 17.70%, giving an observed loss rate of 82.30% on EAD at default of EUR 130,936,042. Collateral coverage of current EAD is 33.67%; the secured EAD share is 33.63%.

Data quality is 99.78% record pass rate across 20 completed batches. 85 error findings and 2 warning findings were recorded; 79 records are retained in quarantine and were not loaded to curated tables. Warnings remain in curated data with their rule evidence, rather than being silently deleted.

Portfolio PD 12M	**1.78%**	Exposure-weighted configured PD.
Observed 12M default rate	**5.17%**	Count-based backward cohort.
Investment-grade share	**46.37%**	EAD rated A or B.
Upgrade / downgrade rate	**10.82% / 13.67%**	Matched 12-month migration.
Recovery rate	**17.70%**	Recoveries / EAD at default.
Loss rate	**82.30%**	Loss / EAD at default.
Collateral coverage	**33.67%**	Haircut-adjusted collateral / current EAD.
Quality score	**99.78%**	Valid / received records.



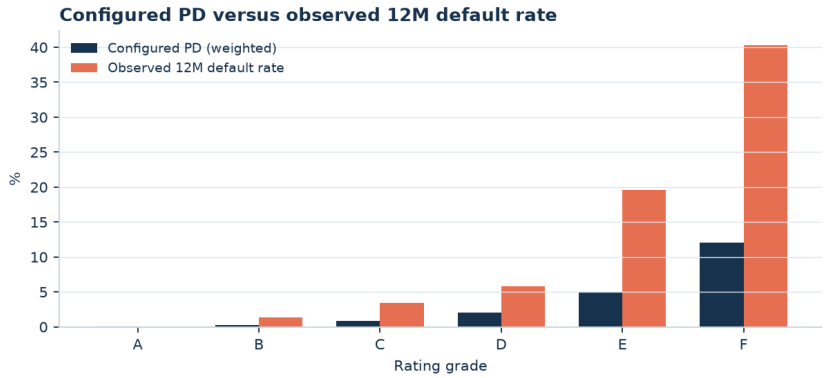


2. Rating & Scoring

Distribution, configured PD 12M, observed default rates and rating migration.

2.1 Rating distribution

Rating	Exposures	EAD (EUR)	Share count	Share EAD	PD 12M	Expected default EAD
A	497	782,499,983	23.72%	25.98%	0.10%	782,500
B	412	614,204,037	19.67%	20.39%	0.30%	1,842,612
C	483	664,404,222	23.05%	22.06%	0.80%	5,315,234
D	352	472,898,146	16.80%	15.70%	2.00%	9,457,963
E	225	301,758,647	10.74%	10.02%	5.00%	15,087,932
F	126	176,245,199	6.01%	5.85%	12.00%	21,149,424



2.2 Observed default rates

The observed default rate uses a 12-month backward cohort. The denominator is the set of performing exposures in the latest quarter snapshot on or before reporting date minus 12 months. A default is counted once per exposure when the first observed default date lies after that observation start and on or before the reporting date. The configured PD is shown separately and is not calibrated to the observed rate in this prototype.

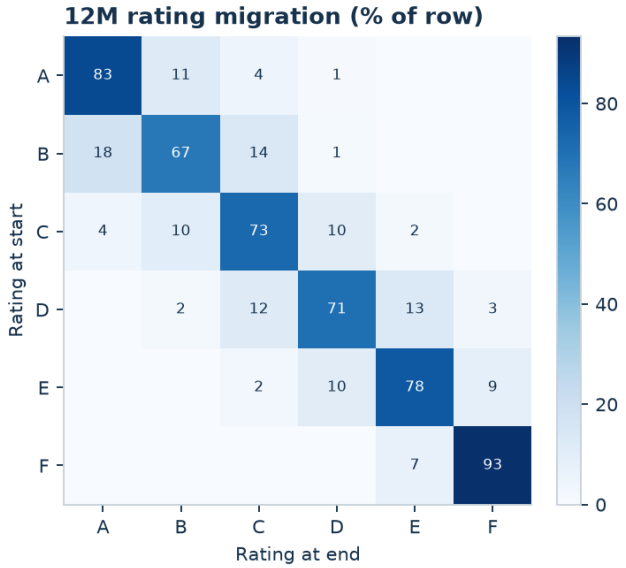
Rating	Base exposures	Base EAD	Defaults	Default EAD	Observed rate	Observed rate (EAD)	Configured PD
A	426	671,769,113	0	0	0.00%	0.00%	0.10%
B	375	586,291,812	5	3,205,203	1.33%	0.55%	0.30%
C	405	507,828,566	14	19,657,199	3.46%	3.87%	0.80%
D	295	415,890,220	17	19,532,244	5.76%	4.70%	2.00%
E	138	210,142,019	27	59,054,007	19.57%	28.10%	5.00%
F	62	80,178,901	25	18,036,953	40.32%	22.50%	12.00%

Bank	Base exposures	Defaults	Observed rate	Observed EAD rate	Configured PD
SSK001	372	9	2.42%	0.65%	1.27%
SSK002	268	11	4.10%	3.14%	1.36%
SSK003	445	46	10.34%	11.84%	2.10%
SSK004	250	2	0.80%	0.25%	0.53%
SSK005	366	20	5.46%	5.44%	1.42%

2.3 Rating migration

Migration compares 2025-09-30 with 2026-09-30. 1,719 exposures are matched. 380 entries and 97 exits are reported separately. Upgrade rate is 10.82%; downgrade rate is 13.67%.

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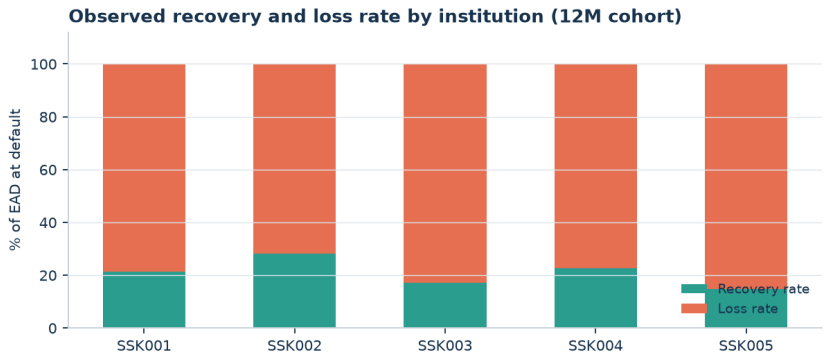
From / to	A	B	C	D	E	F
	83.50%	11.33%	4.43%	0.74%	0.00%	0.00%
	17.51%	67.23%	13.84%	1.41%	0.00%	0.00%
	4.13%	10.08%	73.13%	10.34%	2.07%	0.26%
	0.00%	1.61%	11.61%	70.65%	12.90%	3.23%
	0.00%	0.00%	2.33%	10.47%	78.49%	8.72%
	0.00%	0.00%	0.00%	0.00%	6.67%	93.33%

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

3. Loss estimation

Observed recovery, loss severity, collateral coverage and cashflow timing.

The observed default cohort covers 96 defaults with EAD at default of EUR 130,936,042. The observed loss amount is EUR 107,758,217. Collateral is used only for coverage analysis and is not subtracted again from loss, preventing double counting. Observed recovery rate is 17.70%; observed loss rate is 82.30%. 27.08% of defaults have no observed recovery by the reporting date.

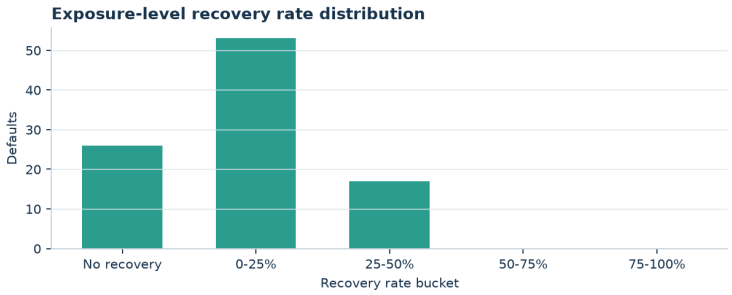


3.1 Recovery and loss by institution

Bank	Defaults	EAD at default	Recoveries	Recovery rate	Loss	Loss rate
SSK001	10	7,829,957	1,674,108	21.38%	6,155,849	78.62%
SSK002	11	10,502,889	2,944,987	28.04%	7,557,903	71.96%
SSK003	50	81,690,776	13,945,364	17.07%	67,745,412	82.93%
SSK004	2	993,083	225,741	22.73%	767,342	77.27%
SSK005	23	29,919,337	4,387,626	14.66%	25,531,711	85.34%

3.2 Recovery rate by rating at default

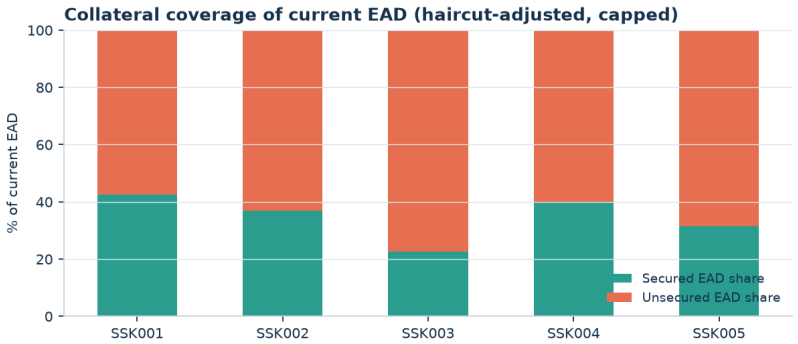
Rating	Defaults	EAD at default	Recoveries	Recovery rate	Loss rate
A	1	3,992,107	900,433	22.56%	77.44%
B	4	2,861,622	885,549	30.95%	69.05%
C	13	16,042,621	3,720,842	23.19%	76.81%
D	18	19,896,706	3,275,875	16.46%	83.54%
E	29	47,166,541	6,116,929	12.97%	87.03%
F	31	40,976,444	8,278,198	20.20%	79.80%



4. Collateral and recovery cashflows

Latest valuations, haircuts, coverage, cashflow timing and cumulative recovery.

Collateral coverage is based on the latest valuation on or before the reporting date. Configured haircuts are 20% for real estate, 5% for financial collateral and 40% for other collateral. Total adjusted collateral is EUR 1,014,010,759; the unsecured EAD amount is EUR 1,998,979,383. 1 latest collateral items exceed the 3-year valuation age warning threshold.



4.1 Collateral by institution

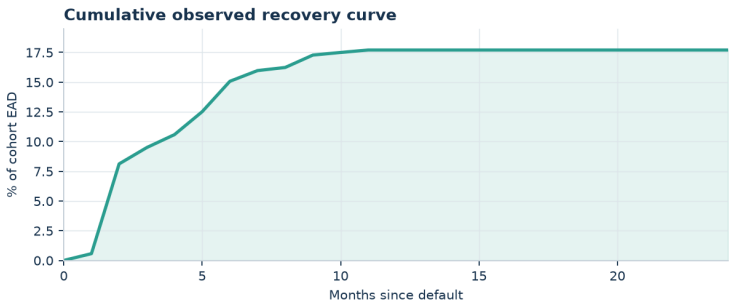
Bank	Exposures	Current EAD	Collateral value	Adjusted value	Secured EAD share	Coverage
SSK001	449	685,934,216	n/a	292,359,279	42.60%	42.62%
SSK002	332	428,820,608	n/a	158,036,266	36.83%	36.85%
SSK003	557	799,603,375	n/a	179,580,660	22.45%	22.46%
SSK004	302	468,017,981	n/a	186,376,281	39.68%	39.82%
SSK005	455	629,634,054	n/a	197,658,274	31.38%	31.39%

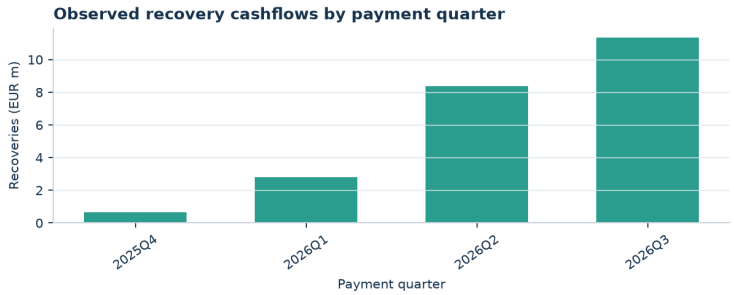
4.2 Collateral type mix

Type	Items	Value	Adjusted	Haircut reduction	Share adjusted
REAL_ESTATE	634	741,513,263	593,210,610	148,302,653	58.50%
OTHER	308	318,203,285	190,921,971	127,281,314	18.83%
FINANCIAL_COLLATERAL	203	241,977,030	229,878,178	12,098,851	22.67%

4.3 Recovery cashflow timing

Payment period	Cashflows	Recovered	Share	Cumulative share
2025Q4	4	637,616	2.75%	2.75%
2026Q1	21	2,801,095	12.09%	14.84%
2026Q2	39	8,383,160	36.17%	51.01%
2026Q3	57	11,355,955	48.99%	100.00%




Observed recovery cashflows by payment quarter
Recoveries (EUR m)
0
2
4
6
8
10
2025Q4
2026Q1
2026Q2
2026Q3
Payment quarter

5. Peer benchmarking

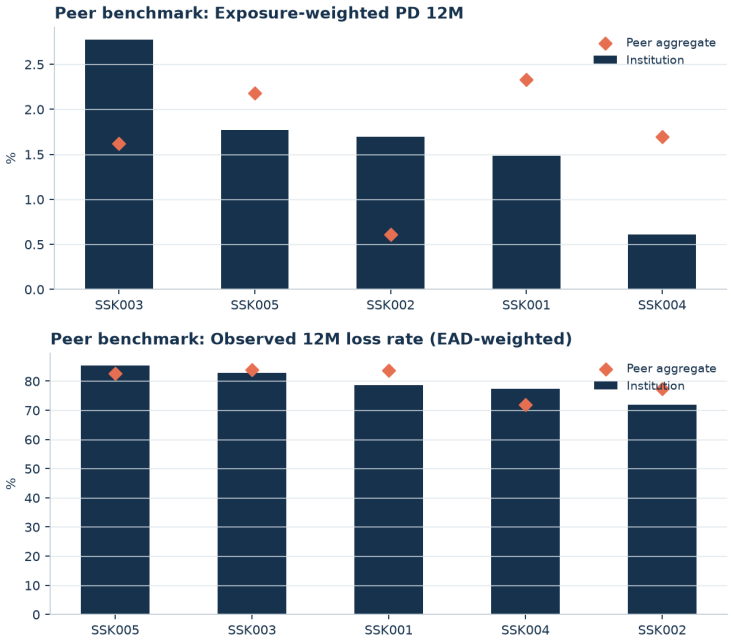
Numerator/denominator peer aggregation, excluding the institution from its own peer reference.

Peer values are calculated from summed numerators and denominators, not by averaging bank-level ratios. The bank under review is excluded from its own peer group. Relative differences above 10% are labelled BETTER or WORSE when the metric has an unambiguous direction; neutral rating-share metrics are informational.

Bank	PD 12M	Default rate	Recovery rate	Loss rate	Coverage
SSK001	1.48%	2.42%	21.38%	78.62%	42.62%
SSK002	1.70%	4.10%	28.04%	71.96%	36.85%
SSK003	2.77%	10.34%	17.07%	82.93%	22.46%
SSK004	0.61%	0.80%	22.73%	77.27%	39.82%
SSK005	1.77%	5.46%	14.66%	85.34%	31.39%

Directional benchmark flags

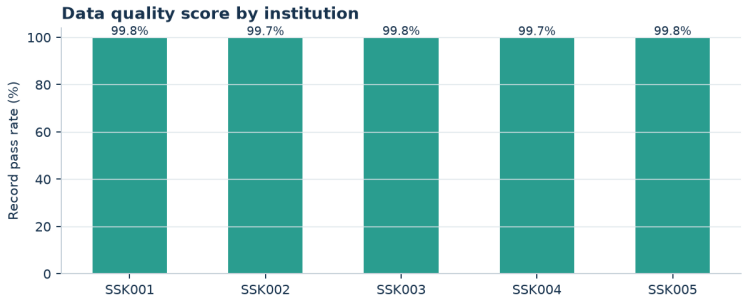
Bank	PD 12M	Default	Recovery	Loss	Coverage
SSK001	BETTER	BETTER	BETTER	IN_LINE	BETTER
SSK002	WORSE	WORSE	BETTER	IN_LINE	IN_LINE
SSK003	WORSE	WORSE	IN_LINE	IN_LINE	WORSE
SSK004	BETTER	BETTER	WORSE	IN_LINE	IN_LINE
SSK005	BETTER	BETTER	WORSE	IN_LINE	IN_LINE



6. Data quality and lineage

Validated records, error quarantine, warnings and pipeline traceability.

The data quality score is 99.78% (valid records / received records). 20 batches completed and 0 duplicate batches were skipped. 85 error findings and 2 warning findings are available in the audit tables. Quarantine holds 79 records; these were not loaded into curated analytical tables. Warnings are retained in curated data with their rule evidence.



6.1 Quality by institution

Bank	Batches	Received	Valid	Rejected	Score	Errors	Warnings
SSK001	4	7,722	7,707	15	9981.00%	16	0
SSK002	4	5,688	5,672	16	9972.00%	16	0
SSK003	4	9,624	9,607	17	9982.00%	19	1
SSK004	4	5,187	5,173	14	9973.00%	15	1
SSK005	4	7,782	7,765	17	9978.00%	18	0

6.2 Top data-quality rules

Rule	Name	Severity	Findings	Records
COMP-EXP-005	Rating grade is required	ERROR	5	5
CONS-EXP-002	Rating grade is on the configured scale	ERROR	5	5
CONS-RAT-003	Rating history exposure exists	ERROR	5	5
PLAUS-COL-001	Collateral value is non-negative	ERROR	5	5
PLAUS-EXP-001	EAD is non-negative	ERROR	5	5
PLAUS-REC-001	Recovery amount is non-negative	ERROR	5	5
UNIQ-COL-001	Collateral valuation is unique within batch	ERROR	5	5
UNIQ-EXP-001	Exposure is unique within the import batch	ERROR	5	5
UNIQ-RAT-001	Rating observation is unique within batch	ERROR	5	5
UNIQ-REC-001	Cashflow ID is unique within batch	ERROR	5	5
UNIQ-REC-003	Recovery business record is not duplicated	ERROR	5	5
COMP-EXP-002	Bank ID is required	ERROR	4	4

6.3 Data lineage and batch audit

Bank	Dataset	Batches	Received	Valid	Rejected	Latest import
SSK001	collateral	1	684	681	3	2026-11-11
SSK001	exposures	1	3,468	3,462	6	2026-11-11
SSK001	rating_history	1	3,469	3,466	3	2026-11-11
SSK001	recovery_cashflows	1	101	98	3	2026-11-11
SSK002	collateral	1	456	452	4	2026-11-11
SSK002	exposures	1	2,567	2,561	6	2026-11-11
SSK002	rating_history	1	2,568	2,565	3	2026-11-11
SSK002	recovery_cashflows	1	97	94	3	2026-11-11
SSK003	collateral	1	547	544	3	2026-11-11
SSK003	exposures	1	4,386	4,379	7	2026-11-11
SSK003	rating_history	1	4,387	4,384	3	2026-11-11

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Bank	Dataset	Batches	Received	Valid	Rejected	Latest import
SSK003	recovery_cashflows	1	304	300	4	2026-11-11
SSK004	collateral	1	525	522	3	2026-11-11
SSK004	exposures	1	2,322	2,317	5	2026-11-11
SSK004	rating_history	1	2,323	2,320	3	2026-11-11
SSK004	recovery_cashflows	1	17	14	3	2026-11-11
SSK005	collateral	1	514	510	4	2026-11-11
SSK005	exposures	1	3,557	3,550	7	2026-11-11
SSK005	rating_history	1	3,558	3,555	3	2026-11-11
SSK005	recovery_cashflows	1	153	150	3	2026-11-11

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7. Methodology and assumptions

Definitions that are deliberately explicit because the prototype is a demonstration.

7.1 Shared data platform

Heterogeneous CSV, Excel and source-database files are mapped into canonical staging entities. Completeness, uniqueness, consistency and plausibility rules run before curated loading. Error records are quarantined with their rule IDs and raw JSON; warnings are retained in curated data. Every batch, quality finding and pipeline run is traceable to a source file hash and import timestamp.

7.2 Rating & Scoring

Rating distribution and expected default are based on the configured, explicitly fictional rating-to-PD mapping: A 0.1%, B 0.3%, C 0.8%, D 2.0%, E 5.0%, F 12.0%. 12-month through-the-cycle-style demonstration PD by rating grade. Values are configurable assumptions, not validated model parameters. Observed default rates use a separate 12-month backward cohort and are not used to calibrate the configured PD in this prototype. Migration compares two snapshots, reports entries and exits separately and uses matched exposures for the transition matrix.

7.3 Loss Estimation

EAD at default is the latest available exposure snapshot on or before the observed default date. Recoveries are cashflows paid after default and on or before the reporting date. Loss is EAD at default minus realised recoveries. No discounting, indirect workout costs or collateral double counting are applied. Collateral coverage uses the latest valuation on or before the reporting date and configured haircuts REAL_ESTATE 20%, FINANCIAL_COLLATERAL 5%, OTHER 40%. Coverage is shown for context, not subtracted again from loss.

7.4 Benchmarking

Benchmark values are calculated from summed numerators and denominators across peer institutions. The institution under review is excluded from its own peer reference. Configured peer groups: PG_URBAN = SSK001, SSK003, SSK005; PG_REGIONAL = SSK002, SSK004. A small peer group means the peer reference is illustrative rather than statistically robust.

7.5 Data quality score

The quality score is the record pass rate: 100 x valid records / received records across completed batches. It is not a measure of model accuracy, calibration or regulatory fitness. The score is computed per batch and aggregated by totals, not by averaging batch percentages.

7.6 Limitations

This is a local DuckDB prototype with synthetic data. It has no real external integrations, no production identity management, no regulatory sign-off and no validated credit-risk models. Peer groups are tiny and illustrative. The configured PDs, haircuts and loss assumptions are demonstration parameters only. A production implementation would require model validation, data-lineage controls, access management, reconciliation, backtesting, audit sign-off and an enterprise database/BI deployment.

7.7 Reproducibility

Synthetic data is generated with a fixed seed and a generation manifest. The pipeline can be rerun with python synthetic_data/generate_data.py followed by python -m etl.pipeline --reset-db. Reports are logged in the report_log table and distribution is simulated locally.

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Appendix A. Report audit

Generated artefact metadata and disclaimer.

Item	Value
Report title	Risk Portfolio Reporting - 2026-09-30
Organisation label	Fictional Sparkassen Reporting Community
Scope	Reporting Community
Peer group	COMMUNITY
Generated at	2026-11-11 00:57:17
Pipeline run	RUN-20261111005418-7B456F2D
Batches completed	20
Rows received	36003
Rows valid	35924
Rows rejected	79
Disclaimer	SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

SYNTHETIC DEMONSTRATION DATA - NOT REGULATORY REPORTING AND NOT REAL SPARKASSEN OR CUSTOMER INFORMATION

The report is generated from synthetic demonstration data. It must not be represented as a real Sparkasse report, a regulatory submission, a validated model output or a customer communication.